

Chapter 3: Observational Studies

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In Chapter 2, randomization gave us exchangeability by design. Many scientific questions, however, have to be answered without assigning treatment.

Definition 0.1 (Observational study). An **observational study** is a study in which the investigators observe and record the treatment, the outcome, and other relevant variables, but do not assign the treatment: who receives which treatment is decided by the individuals themselves or by someone other than the investigators.

Example 0.1 (Watching pedestrians instead of randomizing). Let the treatment A be whether one pedestrian looks up at the sky, and the outcome Y whether a nearby pedestrian then looks up too. If we merely record pairs of passers-by, nobody assigns A , so the study is observational

(Definition 0.1). Any association between A and Y could then reflect a shared cause, such as a sudden noise overhead, a worry that randomizing A would have ruled out.

Much of what we know (evolution, plate tectonics, the fact that hot coffee can burn) comes from observation rather than experiment.

This chapter asks: under what conditions can an observational study support a valid causal inference?

This chapter is based on Hernán and Robins (2020, chap. 3, pp. 27-46).

1 3.1 Identifiability Conditions (pp. 27-29)

Example 1.1 (A risk ratio that mixes benefit and prognosis). Let A be heart transplant and Y death. In an ideal marginally randomized experiment (full adherence, no loss to follow-up), exchangeability holds, and an associational risk ratio of, say, 0.7 would be expected to match the causal risk ratio.

Suppose instead that A was not randomized and that sicker patients were more likely to be transplanted. Suppose also that transplant really does lower the risk of death. Then an associational risk ratio of 1.1 may be a **compromise** between two opposing forces:

- the benefit of transplant, which pulls the ratio below 1;
- the worse prognosis of those who were transplanted, which pushes it above 1.

So an observational association may owe more to who got treated than to what treatment does. Observational analyses commonly act *as though* treatment had been randomized within levels of the measured covariates L , an approximation at best. Their validity then hinges on whether the study really resembles a **conditionally randomized experiment**.

Definition 1.1 (Identifiability conditions). Let A be a treatment, Y an outcome with counterfactual outcomes Y^a , and L a set of measured covariates. The **identifiability conditions** (or identifiability assumptions) are three conditions under which the average causal effect of A on Y can be computed from the distribution of the observed data:

1. **consistency**: each individual's observed outcome is her counterfactual outcome under the treatment she actually received, $Y = Y^A$;
2. **conditional exchangeability**: given only the measured covariates L , the counterfactual outcomes are independent of the treatment actually received, $Y^a \perp\!\!\!\perp A \mid L$ for all a ;
3. **positivity**: $\Pr[A = a \mid L = l] > 0$ for every treatment value a in the causal contrast and every l with $\Pr[L = l] \neq 0$.

When all three hold, the observational study can be analyzed as if it were a conditionally randomized experiment.

Example 1.2 (The identifiability conditions in a heart transplant study). Let A be heart transplant, Y death, and L critical condition at baseline. The three conditions of Definition 1.1 then say:

1. consistency: each patient's recorded death status is the one that goes with the transplant decision she actually had;
2. conditional exchangeability: among patients equally critical, those who were and were not transplanted would have had the same risk of death under either decision;
3. positivity: both critical and noncritical patients had some chance of being transplanted and some chance of not being transplanted.

Consistency comes from Chapter 1, exchangeability from Chapter 2, and positivity from Technical Point 2.3. Sections 3.2 to 3.5 take them one at a time and define each formally.

! Data Alone Are Not Enough

An ideal randomized experiment secures the identifiability conditions (Definition 1.1) through its design. An observational study cannot do so, and there the conditions are *assumptions* brought to the data from outside. A causal conclusion from observational data is only as good as the data **and** those assumptions together.

The identifiability conditions are often heroic, which is why causal inferences from observational data are viewed with suspicion.

For simplicity, this chapter considers only studies in which everyone stays under follow-up and adheres to the assigned treatment; Chapters 8 and 9 relax this.

The book notes that Rubin (1974, 1978) extended Neyman’s theory to observational studies (Hernán and Robins 2020, 28).

1.1 Table 3.1

Table 1: The book’s Table 3.1: critical condition at baseline L (1: yes), heart transplant A , and death Y for 20 individuals (Hernán and Robins 2020, 28).

Name	L	A	Y	Name	L	A	Y
Rheia	0	0	0	Leto	1	0	0
Kronos	0	0	1	Ares	1	1	1
Demeter	0	0	0	Athena	1	1	1
Hades	0	0	0	Hephaestus	1	1	1
Hestia	0	1	0	Aphrodite	1	1	1
Poseidon	0	1	0	Polyphemus	1	1	1
Hera	0	1	0	Persephone	1	1	1
Zeus	0	1	1	Hermes	1	1	0
Artemis	1	0	1	Hebe	1	1	0
Apollo	1	0	1	Dionysus	1	1	0

Table 3.1 contains the same data as Table 2.2. If these data came from an observational study in which the three identifiability conditions (Definition 1.1) held, we would compute the same causal risk ratio as in Chapter 2: 1.

Example 1.3 (Standardized risks in Table 3.1). Assume that the data in Table 3.1 satisfy the identifiability conditions (Definition 1.1) with L as the measured covariate, so that each counterfactual risk equals the standardized risk. Stratum-specific risks (8 individuals with $L = 0$, 12 with $L = 1$):

- $L = 0$: $\Pr[Y = 1 \mid A = 1, L = 0] = 1/4$ and $\Pr[Y = 1 \mid A = 0, L = 0] = 1/4$
- $L = 1$: $\Pr[Y = 1 \mid A = 1, L = 1] = 6/9 = 2/3$ and $\Pr[Y = 1 \mid A = 0, L = 1] = 2/3$

with $\Pr[L = 0] = 8/20 = 0.4$ and $\Pr[L = 1] = 12/20 = 0.6$. Standardizing,

$$\begin{aligned}
 \Pr[Y^{a=1} = 1] &= \Pr[Y = 1 \mid A = 1, L = 0] \Pr[L = 0] + \Pr[Y = 1 \mid A = 1, L = 1] \Pr[L = 1] \\
 &= (1/4)(0.4) + (2/3)(0.6) \\
 &= 0.1 + 0.4 \\
 &= 0.5,
 \end{aligned}$$

$$\begin{aligned}
\Pr[Y^{a=0} = 1] &= \Pr[Y = 1 \mid A = 0, L = 0] \Pr[L = 0] + \Pr[Y = 1 \mid A = 0, L = 1] \Pr[L = 1] \\
&= (1/4)(0.4) + (2/3)(0.6) \\
&= 0.1 + 0.4 \\
&= 0.5,
\end{aligned}$$

so the causal risk ratio is $0.5/0.5 = 1$.

Definition 1.2 (Weak and strong ignorability). Let A be a treatment with counterfactual outcomes Y^a , and L a set of measured covariates. Following Rosenbaum and Rubin (1983), treatment assignment is

- **weakly ignorable** given L when conditional exchangeability $Y^a \perp\!\!\!\perp A \mid L$ (for all a) and positivity both hold;
- **strongly ignorable** given L when positivity holds and the counterfactual outcomes Y^a for all values a , taken jointly, are independent of A given L (full conditional exchangeability, the conditional version of Technical Point 2.1) (Hernán and Robins 2020, 28).

Example 1.4 (Ignorability in Table 3.1). Read Table 3.1 as a conditionally randomized experiment in which a coin with $\Pr[A = 1 \mid L = 1] = 0.75$ and $\Pr[A = 1 \mid L = 0] = 0.50$ assigned treatment. The coin ignores everything about an individual except L , so the pair $\{Y^{a=0}, Y^{a=1}\}$ is independent of A given L , and neither probability is 0 or 1. Treatment assignment is therefore strongly ignorable given L , and hence also weakly ignorable (Definition 1.2).

1.2 Identifiability

Definition 1.3 (Identifiability of an average causal effect). Fix a set of assumptions and an effect measure for an average causal effect, such as the causal risk ratio. The effect is (nonparametrically) **identifiable** under those assumptions if, once the distribution of the observed data is known, the assumptions leave exactly one possible value of the effect measure. It is **nonidentifiable** if more than one value remains possible (Hernán and Robins 2020, Fine Point 3.1, p. 29).

Example 1.5 (Identifiability in Table 3.1). If Table 3.1 came from a conditionally randomized experiment, $Y^a \perp\!\!\!\perp A \mid L$ and positivity hold by design, consistency holds for the interventions the protocol carried out, and the causal risk ratio of 1 is identified with no further assumptions. If it came from an observational study, then even taking consistency as given and with positivity visible in the table, we can conclude that the causal risk ratio is 1 only by adding the *identifying assumption* $Y^a \perp\!\!\!\perp A \mid L$, which is external to the data. Without it, the same data leave the causal risk ratio undetermined (Definition 1.3):

- it can be below 1, if the treated carry more of the other predictors of death besides L ;
- it can be above 1, if the untreated carry more of them;
- it can be 1, if those other predictors are balanced within levels of L , i.e., if $Y^a \perp\!\!\!\perp A \mid L$.

i Fine Point 3.1: Identifiability of causal effects

The book defines identifiability (Definition 1.3) in this Fine Point and illustrates it with Table 3.1 (Example 1.5).

1.3 Other identifiability conditions

Remark 1.1 (Other routes to identification). When any of the three conditions in Definition 1.1 fails, the analogy with a conditionally randomized experiment breaks down. Other approaches rely on different identifiability conditions; for example, **instrumental variable** methods (Chapter 16) assume that a predictor of treatment behaves as if randomly assigned conditional on measured covariates.

Fields where observational studies often resemble conditionally randomized experiments (e.g., epidemiology) have tended to rely on that resemblance, while fields where they seldom do (e.g., economics) have leaned on instrumental variables. Until Chapter 16, the book focuses on the first family.

2 3.2 Exchangeability (pp. 29-31)

Definition 2.1 (Independent predictor of the outcome). Let A be a treatment and Y an outcome. A covariate is an **independent predictor of the outcome** if it is associated with Y within levels of A . When Y is dichotomous, such covariates are commonly called **risk factors**.

Example 2.1 (Critical condition is a risk factor in Table 3.1). In Table 3.1, the risk of death among the untreated is $1/4$ when $L = 0$ and $2/3$ when $L = 1$; among the treated it is $1/4$ when $L = 0$ and $6/9 = 2/3$ when $L = 1$. Critical condition L is associated with death Y within both levels of A , so it is an independent predictor of the outcome, a risk factor (Definition 2.1).

Remark 2.1 (Why marginal randomization gives exchangeability). Let A be a treatment with counterfactual outcomes Y^a . Marginal randomization of A (assignment that ignores every characteristic of the individual) makes the treated and untreated groups exchangeable ($Y^a \perp\!\!\!\perp A$) because it spreads the independent predictors of the outcome (Definition 2.1) equally, in expectation, across the two groups.

Example 2.2 (Marginal exchangeability fails in Table 3.1). In Table 3.1, 69% of the treated (9/13) but only 43% of the untreated (3/7) were in critical condition ($L = 1$). Since L is a risk factor (Example 2.1), this imbalance suggests the treated had a worse prognosis. To check, suppose, as in Chapter 2, that the data come from a conditionally randomized experiment, so that consistency, $Y^a \perp\!\!\!\perp A \mid L$, and positivity hold (positivity holds here because $\Pr[A = 1 \mid L = 1] = 0.75$ and $\Pr[A = 1 \mid L = 0] = 0.50$). Then the risk the treated would have had without treatment is

$$\begin{aligned} \Pr[Y^{a=0} = 1 \mid A = 1] &= \sum_l \Pr[Y = 1 \mid A = 0, L = l] \Pr[L = l \mid A = 1] \\ &= (1/4)(4/13) + (2/3)(9/13) \\ &= 7/13 \\ &\approx 0.54, \end{aligned}$$

while the untreated had $\Pr[Y^{a=0} = 1 \mid A = 0] = \Pr[Y = 1 \mid A = 0] = 3/7 \approx 0.43$. The two differ, so marginal exchangeability $Y^a \perp\!\!\!\perp A$ does not hold.

Definition 2.2 (Conditional exchangeability). Let A be a treatment with counterfactual outcomes Y^a , and L a set of measured covariates. The treated and untreated are **conditionally exchangeable** given L when

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for all } a.$$

Equivalently, for a dichotomous treatment and outcome, $\Pr[Y^a = 1 \mid A = 1, L = l] = \Pr[Y^a = 1 \mid A = 0, L = l]$ for all a and for all l at which both conditional probabilities are defined.

A conditionally randomized experiment produces conditional exchangeability: inside each stratum of L , treatment is assigned at random, so the remaining outcome predictors are balanced across treatment groups.

Example 2.3 (Two descriptions of the Table 3.1 study). Suppose L is the **only** outcome predictor unequally distributed between treatment groups. Then the study in Table 3.1 can be described in two logically equivalent ways:

1. an observational study in which $\Pr[A = 1 \mid L = 1] = 9/12 = 0.75$ and $\Pr[A = 1 \mid L = 0] = 4/8 = 0.50$, e.g., because doctors direct the scarce hearts to those in critical condition; or
2. a conditionally randomized experiment, not blinded, whose investigators gave $A = 1$ with probability 0.75 when $L = 1$ and 0.50 when $L = 0$.

Under either description, $Y^a \perp\!\!\!\perp A \mid L$ holds (Definition 2.2), and, together with consistency and positivity (which holds here, since $\Pr[A = 1 \mid L = l]$ is 0.75 or 0.50), standardization or IP weighting identifies the causal effect.

Chapter 7 calls such outcome predictors *confounders*. Nothing in the data can show whether L is the only imbalanced outcome predictor, so an observational analysis has to assume it.

2.1 Not every imbalanced variable must be in L

Example 2.4 (HLA compatibility). In the heart transplant study (A : transplant, L : critical condition, Y : death), donor hearts go to recipients with compatible **HLA** (human leukocyte antigen) genes, so HLA genes are unequally distributed between treatment groups. Suppose that HLA genes do not predict death once L and A are known. Then, within levels of L , treatment is still effectively random with respect to the outcome, and HLA need not be included in L .

2.2 Unmeasured outcome predictors

Example 2.5 (Doctors who prefer nonsmokers). In the heart transplant study (A : transplant, L : critical condition, Y : death), let U denote smoking, which the investigators did not measure and which predicts death even within levels of L and A . Suppose also that doctors, without the investigators knowing it, favor nonsmokers when allocating hearts. Within the stratum $L = 1$, smokers ($U = 1$) are then less likely to be treated, so smoking is less common among the treated than the untreated, and $Y^a \perp\!\!\!\perp A \mid L$ fails.

Remark 2.2 (Conditional exchangeability cannot be verified). Let A be a treatment, Y a dichotomous outcome with counterfactual outcomes Y^a , L the measured covariates, and U an unmeasured variable. Conditional exchangeability $Y^a \perp\!\!\!\perp A \mid L$ generally fails when some unmeasured variable U both independently predicts the outcome and, within strata of L , changes the chance of being treated. Even when it holds, it cannot be checked: verifying it would require comparing $\Pr[Y^a = 1 \mid A = a, L = l]$ with $\Pr[Y^a = 1 \mid A \neq a, L = l]$ at each l

where both groups exist, but Y^a is unknown for individuals with $A \neq a$.

💡 Use Expert Knowledge to Choose the Covariates

Conditional exchangeability cannot be tested, so investigators make it more plausible by using expert knowledge to choose and measure many relevant covariates, and then assuming exchangeability within their joint strata.

Variables that were never measured cannot enter standardization or IP weighting, so the effect is not identified when the measured L falls short of conditional exchangeability. Collecting data on smoking would not remove the possibility that other unknown outcome predictors remain imbalanced.

No matter how many variables are in L , the assumption cannot be tested. The validity of the inference rests on the investigators' expert knowledge, encoded as conditional exchangeability, which supplements the data.

i Fine Point 3.2: Crossover randomized experiments

Fine Point 2.1 identified individual effects in crossover experiments (periods $t = 0, 1$) under three conditions: (i) no carryover, $Y_{i,t=1}^{a_0,a_1} = Y_{i,t=1}^{a_1}$; (ii) a time-constant individual effect, $Y_{it}^{a_t=1} - Y_{it}^{a_t=0} = \alpha_i$; and (iii) a time-constant untreated outcome, $Y_{it}^{a_t=0} = \beta_i$. Now drop (iii), and randomize the order: each individual gets $(A_{i0}, A_{i1}) = (0, 1)$ or $(1, 0)$ with probability 0.5. Let $r_i \stackrel{\text{def}}{=} Y_{i1}^{a_1=0} - Y_{i0}^{a_0=0}$. Under (i), (ii), and consistency:

$$\begin{aligned} \text{if } (A_{i0}, A_{i1}) = (0, 1) : \quad Y_{i1} - Y_{i0} &= Y_{i1}^{a_1=1} - Y_{i0}^{a_0=0} && \text{(consistency, (i))} \\ &= (\alpha_i + Y_{i1}^{a_1=0}) - Y_{i0}^{a_0=0} && \text{(ii)} \\ &= \alpha_i + r_i, && \text{(definition of } r_i) \\ \text{if } (A_{i0}, A_{i1}) = (1, 0) : \quad Y_{i0} - Y_{i1} &= (\alpha_i + Y_{i0}^{a_0=0}) - Y_{i1}^{a_1=0} && \text{(consistency, (i), (ii))} \\ &= \alpha_i - r_i. && \text{(definition of } r_i) \end{aligned}$$

Individual effects are no longer identified because r_i is unknown. But, using $A_{i0} = 1 - A_{i1}$,

$$\begin{aligned} E[(Y_{i1} - Y_{i0})A_{i1} + (Y_{i0} - Y_{i1})A_{i0}] &= E[(\alpha_i + r_i)A_{i1} + (\alpha_i - r_i)A_{i0}] \\ &= E[\alpha_i(A_{i1} + A_{i0})] + E[r_i(A_{i1} - A_{i0})] \\ &= E[\alpha_i] + E[r_i] E[A_{i1} - A_{i0}] && (A_{i1} + A_{i0} = 1; \text{ randomization: } A \perp\!\!\!\perp r_i) \\ &= E[\alpha_i] + E[r_i](0.5 - 0.5) \\ &= E[\alpha_i]. \end{aligned}$$

If only (i) holds, the same mean estimates $(E[\alpha_{i1}] + E[\alpha_{i0}])/2$, where $\alpha_{it} \stackrel{\text{def}}{=} Y_{it}^{a_t=1} - Y_{it}^{a_t=0}$, i.e., the average of the period-specific average effects. The book concludes that, for the treatments and outcomes it studies, the no-carryover assumption is implausible (Hernán and Robins 2020, Fine Point 3.2, p. 31).

3 3.3 Positivity (pp. 32-33)

An experiment that assigned everyone to $A = 1$ (or everyone to $A = 0$) could not estimate the average causal effect. Treatment must be assigned so that each treatment level has a **positive probability**.

Definition 3.1 (Positivity). Let A be a treatment and L the covariates used to achieve conditional exchangeability (Definition 2.2). **Positivity** holds when

$$\Pr[A = a \mid L = l] > 0 \quad \text{for all values } l \text{ with } \Pr[L = l] \neq 0 \text{ in the population of interest,}$$

for every treatment value a involved in the causal contrast. Positivity is sometimes called the **experimental treatment assumption** (Hernán and Robins 2020, 32).

Example 3.1 (Positivity in Table 3.1). In Table 3.1, $\Pr[A = 1 \mid L = 1] = 9/12 = 0.75$ and $\Pr[A = 1 \mid L = 0] = 4/8 = 0.50$. Neither is 0 or 1, so $\Pr[A = a \mid L = l] > 0$ for both $a \in \{0, 1\}$ and both $l \in \{0, 1\}$, and positivity holds (Definition 3.1).

Randomized experiments guarantee positivity by design, which is why Chapter 2 hardly mentioned it.

3.1 Two refinements

Remark 3.1 (Where positivity is needed). Take the heart transplant study (A : transplant, L : critical condition, Y : death), with the average causal effect of A on Y as the target. Positivity (Definition 3.1) is a requirement on two things only:

- Values l **present in the population of interest**: if the study were restricted to $L = 1$, positivity in $L = 0$ would be irrelevant.
- The variables L **required for exchangeability**. Suppose eye color neither predicts death given L and A nor was used to assign treatment. Then eye color need not be adjusted for, and we do not ask whether people with blue eyes had a positive probability of transplant.

3.2 Positivity in observational studies

Example 3.2 (A positivity violation). In the heart transplant study (A : transplant, L : critical condition), suppose doctors always transplanted a heart to individuals in critical condition. Then $\Pr[A = 0 \mid L = 1] = 0$ and positivity (Definition 3.1) fails (book Figure 3.1). The data would contain no untreated individuals with $L = 1$ who could stand in for what would have happened to the treated with $L = 1$ had they been untreated.

Remark 3.2 (Positivity can sometimes be checked). Let A be a treatment and L the covariates required for exchangeability. Unlike exchangeability, positivity can sometimes be **checked empirically** (Chapter 12): when L has few levels, the data can show that every treatment value occurs at every level of L . That is how it was checked in Table 3.1, where both levels of L contain treated and untreated individuals (Example 3.1).

i Technical Point 3.1: Positivity for standardization and IP weighting

The standardized mean $\sum_l E[Y \mid A = a, L = l] \Pr[L = l]$ is defined only if $E[Y \mid A = a, L = l]$ is defined for every l with $\Pr[L = l] \neq 0$, i.e., only under positivity; otherwise it is undefined. For IP weighting, write $f[a \mid L] \stackrel{\text{def}}{=} \Pr[A = a \mid L]$, and $f[A \mid L]$ for its value at each individual's own treatment A . The version $E\left[\frac{I(A=a)Y}{f[a \mid L]}\right]$ is undefined without positivity (it involves $0/0$). The version $E\left[\frac{I(A=a)Y}{f[A \mid L]}\right]$ is always defined, because $f[A \mid L]$ is never 0, but without positivity it no longer equals the counterfactual mean.

Proposition 3.1 (The IP weighted mean without positivity). *Let A be a discrete treatment, L a covariate taking finitely many values, Y an outcome with finite mean, and $f[a | l] \stackrel{\text{def}}{=} \Pr[A = a | L = l]$; write $f[A | L]$ for this function evaluated at each individual's own values of A and L . For a treatment value a , let $Q(a) \stackrel{\text{def}}{=} \{l : \Pr[L = l] > 0 \text{ and } f[a | l] > 0\}$ be the set of covariate values at which $A = a$ occurs with positive probability, and suppose $\Pr[L \in Q(a)] > 0$. Then*

$$\mathbb{E}\left[\frac{I(A=a)Y}{f[A | L]}\right] = \Pr[L \in Q(a)] \sum_{l \in Q(a)} \mathbb{E}[Y | A = a, L = l] \Pr[L = l | L \in Q(a)]. \quad (1)$$

If, in addition, Y^a has finite mean and consistency ($Y = Y^A$) and conditional exchangeability ($Y^a \perp\!\!\!\perp A | L$) hold, the right-hand side of Equation 1 equals $\mathbb{E}[Y^a | L \in Q(a)] \Pr[L \in Q(a)]$.

i Proof

Proof. Condition on L . For l with $\Pr[L = l] > 0$ but $l \notin Q(a)$, the indicator $I(A = a)$ is 0 with probability 1 given $L = l$, so that stratum contributes 0. For $l \in Q(a)$, $f[A | L] = f[a | l]$ whenever $I(A = a) = 1$, so

$$\begin{aligned} \mathbb{E}\left[\frac{I(A=a)Y}{f[A | L]} \mid L = l\right] &= \frac{\mathbb{E}[I(A=a)Y \mid L = l]}{f[a | l]} \\ &= \frac{\mathbb{E}[Y \mid A = a, L = l]f[a | l]}{f[a | l]} \\ &= \mathbb{E}[Y \mid A = a, L = l]. \end{aligned}$$

Averaging over L ,

$$\begin{aligned} \mathbb{E}\left[\frac{I(A=a)Y}{f[A | L]}\right] &= \sum_{l \in Q(a)} \mathbb{E}[Y \mid A = a, L = l] \Pr[L = l] \\ &= \Pr[L \in Q(a)] \sum_{l \in Q(a)} \mathbb{E}[Y \mid A = a, L = l] \Pr[L = l | L \in Q(a)]. \end{aligned}$$

For the second claim, take $l \in Q(a)$. Consistency gives $\mathbb{E}[Y \mid A = a, L = l] = \mathbb{E}[Y^a \mid A = a, L = l]$, and conditional exchangeability gives $\mathbb{E}[Y^a \mid A = a, L = l] = \mathbb{E}[Y^a \mid L = l]$. Summing $\mathbb{E}[Y^a \mid L = l] \Pr[L = l | L \in Q(a)]$ over $l \in Q(a)$ gives $\mathbb{E}[Y^a \mid L \in Q(a)]$. $\square$

Remark 3.3 (An IP weighted contrast without positivity compares different groups). Let A be binary, and let L , Y , f and $Q(a)$ be as in Proposition 3.1 (L with finitely many values, Y with finite mean); assume also that $Y^{a=0}$ and $Y^{a=1}$ have finite means, $\Pr[L \in Q(0)] > 0$, and $\Pr[L \in Q(1)] > 0$. If positivity (Definition 3.1) fails, some l with $\Pr[L = l] > 0$ has $f[a | l] = 0$ for one value a , and then $f[1 - a | l] = 1$; so that l lies in exactly one of $Q(0)$ and $Q(1)$, and $Q(0) \neq Q(1)$. The IP weighted contrast $\mathbb{E}\left[\frac{I(A=1)Y}{f[A | L]}\right] - \mathbb{E}\left[\frac{I(A=0)Y}{f[A | L]}\right]$ then compares two different groups and has no causal interpretation, even under exchangeability. Under positivity, $Q(0) = Q(1)$ contains every l with $\Pr[L = l] > 0$, so $\Pr[L \in Q(a)] = 1$, and under consistency and exchangeability the contrast equals the average causal effect $\mathbb{E}[Y^{a=1}] - \mathbb{E}[Y^{a=0}]$.

4 3.4 Consistency: First, Define the Counterfactual Outcome (pp. 33-38)

Definition 4.1 (Consistency). Let A be a treatment, Y the observed outcome, and Y^a the counterfactual outcome under treatment value a . **Consistency** holds when, for every individual, the outcome we observe is the counterfactual outcome for the treatment value she actually got: $Y = Y^{a=1}$ for those with $A = 1$ and $Y = Y^{a=0}$ for those with $A = 0$. In one line,

$$Y = Y^A,$$

where Y^A means Y^a with a set to the individual's actual treatment A .

Example 4.1 (Aspirin and death). Let A be taking aspirin and Y death. For someone with $A = 1$ and $Y = 1$, consistency (Definition 4.1) gives $Y^{a=1} = 1$: under the aspirin she in fact took, she dies. Here the condition looks too obvious to state, but the rest of this section and Section 3.5 describe settings in which it can fail.

Remark 4.1 (Counterfactuals as primitives). Let A be a treatment, Y the observed outcome, and Y^a the counterfactual outcome under treatment value a . The book takes the counterfactuals Y^a as **primitives**; the observed Y is a function of them. For binary A :

$$Y^A = AY^{a=1} + (1 - A)Y^{a=0},$$

which equals $Y^{a=1}$ when $A = 1$ (the second term vanishes) and $Y^{a=0}$ when $A = 0$ (the first term vanishes).

Remark 4.2 (The two components of consistency). Let A be a treatment, Y the observed outcome, and Y^a the counterfactual outcome under treatment value a . Consistency (Definition 4.1) has two components:

1. pinning down what Y^a means, by saying precisely which treatment the superscript a stands for (this section);
2. connecting those counterfactual outcomes to the outcomes that were actually recorded (Section 3.5).

4.1 Well-defined interventions

Remark 4.3 (Average effects need well-defined individual effects). Let A be a binary treatment and Y a dichotomous outcome. An individual's causal effect $Y^{a=1} - Y^{a=0}$ can be well-defined only if her Y^a is well-defined for both $a = 1$ and $a = 0$. In turn, the average causal effect $\Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1]$ in a given population can be well-defined only if every member of that population has a well-defined individual causal effect.

How do we know the counterfactuals are well-defined? One answer is to tie each treatment value to an intervention, which suffices for Y^a to be well-defined.

Definition 4.2 (Well-defined intervention). Call a treatment value a a **well-defined intervention** if an experimental protocol could spell out the action it stands for. For such an a , Y^a needs no further explanation: it is the outcome under that action.

Example 4.2 (Two heart transplant trials). Two ideal randomized trials of heart transplant ($a = 1$) versus medical therapy ($a = 0$) in the same population:

- **Trial 1** writes a detailed protocol: $a = 1$ means assignment to transplant ($a_0 = 1$) followed by specified preoperative procedures (a_1), anesthesia (a_2), and surgical technique (a_3).

- **Trial 2** deliberately leaves the later steps open: $a = 1$ means assignment to transplant ($a_0 = 1$) followed by whatever happens in routine care, i.e., $a_1 = A_1$, $a_2 = A_2$, $a_3 = A_3$. In both trials $a = 1$ is a well-defined intervention (Definition 4.2), so $Y^{a=1}$ is well-defined. But the two counterfactuals, and hence the two causal effects, will likely differ.

Definition 4.3 (Pragmatic trial). As used in this chapter, a **pragmatic trial** is a randomized trial whose investigators deliberately leave the components of care after assignment unspecified, so that those components take the values they would take in routine practice. Trial 2 of Example 4.2 is a pragmatic trial; Trial 1 is not, because its protocol also fixes a_1 , a_2 , and a_3 .

Formally, $Y^{a=1}$ is the joint counterfactual $Y^{a_0=1, a_1, a_2, a_3}$ in Trial 1 and $Y^{a_0=1, A_1^{a_0=1}, A_2^{a_0=1}, A_3^{a_0=1}}$ in Trial 2; for those assigned to transplant in Trial 2, this equals the observed Y (Technical Point 3.2).

One label, “heart transplant”, covers two different interventions here. Which effect is preferable is a question for the consumers of the research. Trial 1’s effect may vary less across populations, because its components are fixed, whereas the natural values (A_1, A_2, A_3) in Trial 2 depend on the setting; so Trial 1’s result may be easier to **transport** (Chapter 4). This is a question of transportability, not of whether each effect is well-defined.

4.2 Interventions are never perfectly specified

Example 4.3 (The surgeon’s training). Even Trial 1 (Example 4.2) did not specify the surgeon’s training. Because surgical experience affects post-transplant mortality, $\Pr[Y^{a=1} = 1]$ depends on the (unknown) mix of surgeons. Had the protocol specified training, some other detail would still have been left unspecified or open to interpretation.

Remark 4.4 (What imperfect specification implies). Under Trial 1’s protocol, consider the average causal effect $\Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1]$ of heart transplant (versus medical therapy) on death (Example 4.2, Example 4.3).

- This effect can differ across populations that follow the same protocol: a community whose surgeons differ in experience from the trial’s will typically see a different effect.
- Sharper definitions of the interventions generally make it easier to transport the result, that is, to carry the effect over to another population (Chapter 4).
- Components with no effect on the outcome (the color of the surgeons’ scrubs) cannot affect transportability.

Holland’s (1986) slogan “no causation without manipulation” expresses the view that causal inference must rest on well-defined interventions. Such interventions need not be feasible at the time: experts regarded the effects of gene variants on disease as well-defined long before anyone could edit genes.

4.3 Sufficiently well-defined interventions

In experiments, the protocol makes the interventions explicit. In observational studies, two investigators who both speak of “the effect of heart transplant” without defining it may have different interventions in mind, so Y^a stays ill-defined until the experts settle on one intervention at a time.

Definition 4.4 (Sufficiently well-defined intervention). An intervention a is **sufficiently well-defined** when, in the judgment of experts given current knowledge, whatever vagueness is left in the counterfactual outcomes Y^a does not matter for any practical purpose.

Example 4.4 (Settling on one version of heart transplant). Suppose the investigators of an observational study agree to mean Trial 1’s version of heart transplant (Example 4.2). Details remain open even so: the color of the surgeons’ scrubs, which experts agree would not change outcomes, and the surgeons’ training, which affects mortality (Example 4.3). Whether this version of $a = 1$ is sufficiently well-defined (Definition 4.4) depends on whether the experts judge the open details to matter for the question at hand; leaving scrub color open raises no concern, whereas leaving training open is acceptable only if the experts judge that its effect does not matter for the decision the study is meant to inform.

⚠ No Test for Sufficiently Well-Defined

How do we know no meaningful vagueness remains? We don’t. It is a matter of agreement among experts, based on the substantive knowledge available at a particular time, and experts who agree today may judge differently once new knowledge arrives.

i Fine Point 3.3: Possible worlds

Philosophers (Stalnaker 1968, Lewis 1973) defined Y^a as the value of Y in the closest possible world in which the individual received a . Since the closest world to the actual world is itself, $Y^a = Y$ when $A = a$, so consistency always holds under this definition. When $A \neq a$, the closest possible world is vague. Robins and Greenland (2000) proposed replacing closest possible worlds with well-defined interventions, because in observational studies counterfactuals are vague to the degree that the hypothetical interventions are left imprecise.

4.4 Non-intervention variables

Definition 4.5 (Variables that are not interventions). A **non-intervention variable** is a treatment variable A whose values do not correspond to interventions that anyone actually carries out on A itself; typically A can be changed only by intervening on its causes. Whether a particular variable counts as one is a matter of degree, and experts may disagree about it.

Example 4.5 (Obesity and blood pressure). Body weight, blood pressure and cholesterol, and social factors such as socioeconomic status, are non-intervention variables (Definition 4.5).

- A phrase such as “the effect of obesity on heart attack” has no clear meaning, because the counterfactual it refers to is ill-defined. Specifying the timing and the method of weight change (medication, surgery, diet, exercise) would turn it into the effect of *those* interventions, not of obesity.
- Yet experts agree that “blood pressure causes stroke”, by synthesizing physics, in vitro studies, animal experiments, autopsies, and intervention studies.

Whether an effect is ill-defined also depends on the outcome. For obesity and job discrimination (whether an employer invites an applicant to interview after reviewing the application and photograph), the treatment is really whether the employer perceives the applicant as obese, and how the applicant became obese may be irrelevant. A margin note points to Hernán and Taubman (2008), a paper about the difficulties a tyrannical monarch and an inept head of state run into when weighing “the effect of obesity” (Hernán and Robins 2020, 37).

How, then, can “blood pressure A affects stroke Y ” mean anything for a given person? We can take it to mean that some way of changing her blood pressure exists, known or not, that would leave her stroke risk alone apart from the change in blood pressure itself.

Definition 4.6 (Effective no-direct-effect (ENDE) intervention). Let A be a non-intervention variable (Definition 4.5) and Y an outcome. For a given individual, an intervention X (possibly not yet discovered) is an **effective no-direct-effect intervention for a** , written **ENDE(a)**, if it can change that individual's value of A to a and affects her outcome Y only through A , with no (direct) effect of its own. Write $Y^{x,a}$ for the joint counterfactual outcome had X been set to x and A to a . Saying that such an intervention exists amounts to saying that Y^a and $Y^{x,a}$ are both well-defined and equal.

Example 4.6 (Which blood pressure interventions could be ENDE). Let A be blood pressure and Y stroke. Antihypertensive drugs, exercise, and diet all change A . Some antihypertensive drugs might serve as ENDE interventions X (Definition 4.6); diet and exercise cannot, because they affect stroke risk through pathways other than blood pressure.

“ X has no direct effect on Y except through A ” means the same as “the effect of X on Y is completely mediated by A ” (Chapter 23).

Belief in ENDE interventions is easier for variables whose mechanisms are well understood (blood pressure and stroke) than for others (socioeconomic status and myocardial infarction). Where experts disagree, a study design might support the claim that the effect of X on Y is completely mediated by A (Fine Point 6.4).

Blood pressure is really a vector (A_{sys}, A_{dia}) . Accounting for this, the claim that X has no direct effect on Y except through A asserts that $Y^{a_{sys}, a_{dia}}$ and $Y^{x, a_{sys}, a_{dia}}$ are well-defined and equal.

4.5 Average effects of non-interventions

Remark 4.5 (When the average effect of a non-intervention is defined). Let A be a non-intervention variable (Definition 4.5), Y an outcome, and a, a' two values of A , with target the average causal effect $E[Y^a - Y^{a'}]$ in a given population. Because an average causal effect requires a well-defined individual effect for every member of the population (Remark 4.3), this effect is well-defined if every individual in the population has both an ENDE(a) and an ENDE(a') intervention (Definition 4.6).

- If, for some individuals, no ENDE intervention can move A to a and to a' , the population effect is not defined.
- A well-defined population effect therefore requires restricting attention to those individuals who have ENDE interventions available.

Example 4.7 (Small changes in blood pressure). Let A be blood pressure and Y stroke, as in Example 4.6. Considering **small changes** in A makes ENDE interventions more credible. For a change in blood pressure of Δ mm Hg with Δ near 0:

- it is more plausible that (essentially) every individual has an effective intervention;
- direct effects may appear only with large changes (an antihypertensive drug without direct effects at low doses may be toxic at high doses).

Let Y^Δ denote Y^a evaluated at $a = A - \Delta$, where A is each individual's own pre-intervention blood pressure. If (essentially) every individual has an ENDE intervention for changes of size Δ near 0, then Y^Δ is (essentially) well-defined for every individual, and so is the average effect $E[Y^\Delta - Y^{\Delta=0}]$.

i Fine Point 3.4: An interventionist approach to causal inference

The framework of this chapter, sometimes called “interventionist”, builds causal effects from counterfactual outcomes and counterfactual outcomes from well-defined interventions. Because many published causal claims concern non-interventions, the book extends it with conditions,

such as the existence of ENDE interventions (Definition 4.6), under which such claims have a quantitative meaning. The authors do not claim it is the only valid view of causality, but say they know of no other framework whose causal estimates have a practical interpretation (Hernán and Robins 2020, Fine Point 3.4, p. 39).

5 3.5 Consistency: Second, Link Counterfactuals to the Observed Data (pp. 38-40)

Remark 5.1 (When the link holds). Let A be a treatment, Y the observed outcome, and Y^a the counterfactual outcome under treatment value a . The second component of consistency (Remark 4.2) is the “=” in $Y^a = Y$ for individuals with $A = a$.

- **Experiments:** when the intervention a was implemented per protocol, $Y = Y^A = Y^a$ for those with $A = a$ by definition.
- **Observational studies with real-world interventions:** if eligible patients were offered either heart transplant ($A = 1$) or medical therapy ($A = 0$) as part of their care at a particular time and place, then $a = 1$ means whichever procedures came after a transplant offer in that setting, and $Y^{a=1} = Y$ for those with $A = 1$.

In the observational case, the targeted effect is of questionable relevance elsewhere if investigators cannot roughly characterize what “heart transplant” meant in that setting. Madenci et al. (2024) discuss the practical problems of studying heart transplant with observational healthcare databases (Hernán and Robins 2020, 39).

5.1 The choice of intervention depends on the data

Example 5.1 (A well-defined intervention the data cannot link to). Investigators define $a = 1$ as heart transplant with specified preoperative procedures, anesthesia, surgical technique, postoperative care, and immunosuppressive therapy. The data contain only an indicator B of whether a person had a transplant. For an individual with $B = 1$, the well-defined $Y^{a=1}$ need not equal the observed Y .

The book uses a different letter (B) for the recorded variable precisely because the same letter is reserved for cases in which $A = a$ corresponds to the well-defined intervention a .

A Well-Defined Intervention Is Not Enough

- A well-defined intervention (Definition 4.2) that we can describe is **not sufficient** for consistency; we must also be able to link Y^a to the observed Y .
- Conversely, recording a variable A does not make “the effect of A ” meaningful, because the corresponding interventions may be ill-defined.

A well-defined intervention a that nobody in the population received ($A = a$ for no one) amounts to an extreme failure of positivity.

Achieving consistency can be hard in observational studies. A good practice is to make the reasoning transparent so others can challenge it; the target trial (Section 3.6) is a procedure for doing so.

Definition 5.1 (Attributable fraction). Let A be a dichotomous treatment and Y a dichotomous outcome. An **attributable fraction** is the proportion of the observed cases ($Y = 1$) that can be ascribed to treatment. Unlike the causal risk ratio and risk difference, which

compare two counterfactual risks, it is anchored on the cases actually observed. The next two definitions give two versions of it.

Definition 5.2 (Excess fraction). Let A be a dichotomous treatment and Y a dichotomous outcome with $\Pr[Y = 1] > 0$, and let $Y^{a=0}$ be the counterfactual outcome under no treatment. The **excess fraction**, one version of the attributable fraction (Definition 5.1), is the net excess of the observed risk over the risk had everyone been untreated, as a share of the observed risk:

$$\frac{\Pr[Y = 1] - \Pr[Y^{a=0} = 1]}{\Pr[Y = 1]}.$$

Example 5.2 (The ambrosia dinner). All 20 individuals attended a dinner: 10 ate ambrosia ($A = 1$), 10 ate nectar ($A = 0$). The next day 7 of the 10 ambrosia eaters and 1 of the 10 nectar drinkers were sick. Assuming exchangeability,

- causal risk ratio = $0.7/0.1 = 7$, causal risk difference = $0.7 - 0.1 = 0.6$;
- observed risk $\Pr[Y = 1] = 8/20 = 0.4$, versus $\Pr[Y^{a=0} = 1] = 0.1$ had everyone received $a = 0$.

The excess fraction (Definition 5.2) is

$$\begin{aligned} \frac{\Pr[Y = 1] - \Pr[Y^{a=0} = 1]}{\Pr[Y = 1]} &= \frac{0.4 - 0.1}{0.4} \\ &= \frac{0.3}{0.4} \\ &= 0.75. \end{aligned}$$

Had everyone received $a = 0$, we would expect $0.1 \times 20 = 2$ cases rather than the 8 observed; the net excess of 6 cases is 75% of the observed cases, which in this net sense are attributable to ambrosia.

Definition 5.3 (Etiologic fraction). Let A be a dichotomous treatment and Y a dichotomous outcome. The **etiologic fraction** is a second kind of attributable fraction: the proportion of observed cases ($Y = 1$) that treatment mechanically caused (Hernán and Robins 2020, Fine Point 3.5, p. 40). It is a statement about the causal mechanism behind each case, not about the gap between two risks.

Example 5.3 (Excess and etiologic fractions differ at the ambrosia dinner). Keep the ambrosia data (Example 5.2), with exchangeability of ambrosia eaters and nectar drinkers, and fill in each individual's two counterfactual outcomes as follows:

- among the 10 ambrosia eaters, the 7 who fell sick all have $Y^{a=0} = 0$, and exactly one of the other 3 has $Y^{a=0} = 1$;
- among the 10 nectar drinkers, the 1 who fell sick has $Y^{a=1} = 0$, and 7 of the other 9 have $Y^{a=1} = 1$.

These choices agree with the counterfactual risks used before, $\Pr[Y^{a=1} = 1] = (7 + 7)/20 = 0.7$ and $\Pr[Y^{a=0} = 1] = (1 + 1)/20 = 0.1$, and keep exchangeability: $Y^{a=1}$ is 1 for 7 of 10 in each group, and $Y^{a=0}$ is 1 for 1 of 10 in each group.

This individual-level scenario is hypothetical: the data cannot reveal it. Here we count an ambrosia eater's case as caused by ambrosia when that person would have stayed well without it ($Y^{a=0} = 0$). Each of the 7 cases among ambrosia eaters would not have occurred without ambrosia, so ambrosia caused all 7; the single case among nectar drinkers did not involve ambrosia. In this scenario, then, the etiologic fraction (Definition 5.3) is $7/8 = 0.875$. The excess fraction stays at 0.75, a net count: 8 observed cases minus the 2 expected without ambrosia, divided by 8, gives $6/8$. Counting case by case gives $7/8$, so here the excess fraction understates the etiologic fraction, because the ambrosia eater who stayed well but would have fallen sick without ambrosia (the one with $Y^{a=0} = 1$ among the other 3) offsets one of the

seven cases ambrosia caused.

i Fine Point 3.5: Attributable fraction

The book’s Fine Point 3.5 introduces both attributable fractions (Definition 5.1) with the ambrosia dinner (Example 5.2, Example 5.3). Under consistency, the excess fraction needs only the observed risk and one counterfactual risk, so observational data can estimate it under the identifiability conditions. The etiologic fraction is different: it does not rest on excess cases, so it can be computed only in randomized experiments, and even then only under strong assumptions (Greenland and Robins 1988; Robins and Greenland 1989). Fine Point 5.4 revisits the excess fraction in the sufficient-component-cause framework.

6 3.6 The Target Trial (pp. 41-46)

Assuming the three identifiability conditions amounts to viewing an observational analysis as an attempt to **emulate** a hypothetical randomized experiment.

Definition 6.1 (Target trial). The **target trial** (or target experiment) is the hypothetical randomized experiment that would quantify the causal effect of interest. For a given causal effect, the investigator (i) writes the protocol of the target trial, a trial that would be run if it could be, and (ii) explains how the observational data will stand in for that trial.

Example 6.1 (A target trial for aspirin prescription). Take an observational database in which A indicates a prescription of aspirin and Y indicates stroke, restricted to people for whom aspirin is neither clearly required nor clearly ruled out (no absolute indications or contraindications). The target is the average causal effect $\Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1]$. Most experts would accept “prescribed” and “not prescribed” as sufficiently well-defined interventions (Definition 4.4), so a randomized trial of prescribing aspirin *could* have been run. That trial is the target trial (Definition 6.1), and writing down its protocol pins down the causal effect the database is meant to estimate.

If the emulation were successful, the observational study and the target trial (had it been conducted) would give the same results.

The target trial idea, under this and other names, goes back decades: the book cites Dorn (1953), Wold (1954), Cochran (1972), Rubin (1974), Feinstein (1971), and Dawid (2000), and Robins (1986) generalized it to time-varying treatments. The key components were specified by Hernán and Robins (2016) and Hernán et al. (2025). Chapter 22 returns to the framework.

6.1 Key components of the protocol

Algorithm 6.1 (Specifying and emulating a target trial).

1. *Specify the protocol of the target trial (Definition 6.1), with its key components:*
 - *eligibility criteria*
 - *interventions (or, generally, treatment strategies)*
 - *assignment*
 - *outcomes*
 - *start and end of follow-up*
 - *causal contrasts*
2. *For each component, describe how the observational data will emulate it: who in the data is eligible, which intervention each eligible individual followed, and how outcomes are*

ascertained.

3. If the analysis uses standardization or IP weighting, identify the adjustment variables L that are expected to make the treatment groups conditionally exchangeable (Definition 2.2).

Eligibility criteria must be chosen so that every eligible individual could in principle receive every treatment of interest; otherwise some individual causal effects, and hence the average effect, would be undefined. In Parts I and II, every individual becomes eligible at a single time, and that time is **time zero** (the start of follow-up). The acronym PICO (Population, Intervention, Comparator, Outcome; Richardson et al. 1995) summarizes some of these components.

6.2 Interventions and target trials

Remark 6.1 (The target trial for experiments, interventions, and non-interventions). For a treatment variable A and outcome Y , the target trial (Definition 6.1) depends on what kind of variable A is:

- **Experiments:** in a fully adherent trial of low-dose aspirin A and stroke Y , A is the investigators' action, so Y^a is well-defined; the target trial is the trial actually conducted.
- **Observational studies of interventions:** when each value of A is a sufficiently well-defined intervention that some agent could have applied, a target trial *could* have been conducted, and specifying its protocol characterizes the causal effect (Example 6.1).
- **Non-interventions:** the target trial randomizes an ENDE intervention X (Definition 4.6) rather than A , if experts believe one exists.

Whether a variable corresponds to an intervention is often debatable: assignment in a randomized trial uncontroversially is one, socioeconomic status uncontroversially is not, and there is a gradient in between.

6.3 Three non-intervention variables

Example 6.2 (Genotype, blood pressure, and socioeconomic status). Let Y be an outcome of interest, and let A be, in turn, genotype, systolic blood pressure, and socioeconomic status.

1. **Genotype.** Y^a is well-defined if ENDE interventions X (Definition 4.6) could set each target-population member's genotype to a at conception, so that Y^a and $Y^{x,a}$ are equal for each of them. The interventions need not be known: decades ago most experts considered the effect of genotype well-defined, and gene editing was eventually developed.
2. **Systolic blood pressure.** Y^a is well-defined for everyone in the population if ENDE interventions X could bring each individual's blood pressure to a . Many experts believe some drugs act as ENDE interventions in some populations, perhaps only over a limited range, so the effect of *modest* changes can be defined through a target trial that randomizes X (the small-change argument of Example 4.7).
3. **Socioeconomic status.** Most experts doubt that interventions could place people at a given status without affecting Y through other means (they may even disagree on what the term means), so no target trial can be specified and the effect is not well-defined.

Remark 6.2 (Summary of the target trial for interventions and non-interventions). If A is an intervention, the target trial randomizes eligible individuals to values a . If A is a non-intervention and we believe ENDE interventions X exist, the target trial randomizes eligible individuals to an X that sets A to a , and the effect of A is equated with the effect of X . Without that belief, Y^a and the effect of A are not well-defined.

i Fine Point 3.6: When causal inference for non-interventions goes unquestioned

In a trial with assignment Z and received treatment A , the intention-to-treat effect of Z is well-defined because Z is an intervention (Hernán and Robins 2020, Fine Point 3.6, p. 43). With perfect adherence and no direct effect of Z , $Z = A$ and Z is an ENDE intervention for A . Without perfect adherence, Z is not an ENDE intervention for everyone: even with no direct effect of Z on Y , assignment fails to move some individuals' A to the assigned value. The effect of A in the whole population is then well-defined only if some ENDE intervention X on A exists (e.g., adding aspirin to food without the participants' knowledge). Most experts find this so self-evident that “the effect of received treatment” is discussed without mentioning X . Whether or not such an X exists, Z is effective for one subset: individuals who would have $A = 1$ under $Z = 1$ and $A = 0$ under $Z = 0$. For them, Z is an ENDE intervention, so the effect of A in that subset is well-defined and equals the effect of Z . Chapter 16 calls this subset the **compliers**; its members cannot be identified individually, but the effect of A among them can be identified under additional assumptions.

If Z has direct effects on Y not through A , the effect of Z differs from that of X even under full adherence.

Some authors treat “the causal effect of A on Y ” as well-defined even where many experts would not grant that any ENDE intervention X on A exists (Pearl 2009; Schwartz et al. 2016; Glymour and Spiegelman 2016) (Hernán and Robins 2020, 42).

6.4 What a valid emulation needs

Remark 6.3 (Requirements for a valid emulation). Once the protocol of a target trial (Definition 6.1) for the effect of a treatment A on an outcome Y is specified, emulating it with observational data requires

- enough information to map the protocol to the data: identify eligible individuals, classify them by intervention, and ascertain outcomes (step 2 of alg. 6.1);
- when using IP weighting or standardization, enough adjustment variables L for conditional exchangeability (step 3). For “the effect of blood pressure on stroke”, investigators who believe in an ENDE intervention would adjust for variables, such as diet and exercise, that they believe affect both blood pressure and stroke.

Chapter 16 considers alternative identifying conditions for emulating a target trial.

6.5 Anchoring analyses to realistic interventions

Example 6.3 (Body mass index and death). Comparing the risk of death between people with body mass index (BMI) 25 versus 30 at the start of follow-up implies a target trial of an instantaneous, possibly very large, weight change, for which no ENDE intervention is known. A modified analysis could emulate a more reasonable target trial, e.g., one assigning some individuals to a 5% reduction in BMI every year, starting at age 40, for as long as their BMI stays over 25 (a sustained strategy of the kind studied in Part III).

💡 Let the Target Trial Expose Unrealistic Interventions

A written target trial makes the intervention behind an analysis explicit. Decision makers can then see when an analysis, like the BMI comparison in Example 6.3, quietly asks about an intervention nobody could carry out.

Experts may still disagree that ENDE interventions exist for gradual weight change, but such interventions are more plausible than instantaneous weight loss. One course of action is to proceed

as if they existed, adjusting for prognostic factors L , while accepting that the results lack a causal interpretation and may be inadequate for decisions. Such an analysis may still tell us something about what ENDE interventions would do were they to exist, or inspire technologies to create them. Danaei et al. (2016) considered unknown interventions producing progressive weight loss over several years in a study of weight loss and heart disease.

i Fine Point 3.7: Some limits of target trial emulation

“The effect of heart transplant” can mean the average counterfactual outcome of the n eligible individuals under:

- (i) an intervention in which all n individuals receive treatment concurrently; or
- (ii) an intervention in which each individual i receives treatment while everyone else receives the treatment they actually received.

Interpretation (ii), an average of n interventions, is the one relevant to a physician and patient deciding on transplant, and the one the book uses implicitly. Interpretation (i) is not well-defined: it does not say how the health system would be redesigned to supply the hearts and the capacity. If that redesign were specified, current observational data would be inadequate, because they were generated under the old system (e.g., a system that transplants everyone might accept lower-quality organs). The question is close to interference (Fine Point 1.1), but work on interference usually takes interpretation (i) to be well-defined. So observational data may not be enough to describe what would happen if an intervention were scaled up to a whole system.

i Technical Point 3.2: Recursive substitution

For chronologically ordered variables L, A, M, Y with well-defined interventions on L, A, M , the **one-step-ahead counterfactuals** are $L, A^l, M^{l,a}, Y^{l,a,m}$. All other factuals and well-defined counterfactuals are functions of them via recursive substitution, e.g., $A = A^L$, $M^a = M^{L,a}$, $Y^a = Y^{L,a,M^a}$, and $Y = Y^{L,A,M}$. Applied to the two transplant trials, recursive substitution shows why Trial 2’s outcome distribution is harder to transport: it changes whenever the distribution of the natural values $A_1^{a_0=1}, A_2^{a_0=1}, A_3^{a_0=1}$ differs between populations. The argument applies equally to observational and randomized studies.

7 Summary

The identifiability conditions (Definition 1.1) under which an observational study can be analyzed like a conditionally randomized experiment:

1. **Exchangeability** (Definition 2.2): $Y^a \perp\!\!\!\perp A \mid L$ (untestable, so it rests on expert knowledge)
2. **Positivity** (Definition 3.1): $\Pr[A = a \mid L = l] > 0$ for all a in the contrast and all l with $\Pr[L = l] \neq 0$ (sometimes checkable)
3. **Consistency** (Definition 4.1): $Y = Y^A$, which requires (i) sufficiently well-defined interventions and (ii) a link between those interventions and the observed data

Specifying the **target trial** (Definition 6.1) makes the causal question, and what the data must supply to emulate it, explicit. For non-intervention variables, the target trial randomizes an ENDE intervention X (Definition 4.6).

Looking ahead:

- Chapter 4 discusses effect modification and the dependence of causal effects on the population
- Chapter 6 introduces causal diagrams; Chapters 7-9 cover confounding, selection bias, and measurement bias
- Chapter 16 describes instrumental variable methods, which rely on different identifiability conditions
- Chapter 22 returns to target trial emulation

8 References

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